

DISTRIBUTIONS OF RANDOM VARIABLES WITH RANDOM PARAMETERS

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1. Introduction and summary.

Statistical distributions involving random parameters occur in a variety of problems. A number of authors investigated the distribution of stochastic variables where one or more parameters have a priori distributions. Con Seal (1952) in his discussion of compound Poisson Processes, considered the unconditional distribution of a Poisson variate where the apriori distribution of the parameter is taken to be Poisson, Pascal, Pólya or Pearson type III. Domb (1952) investigated the classical problem of finding the probability that m cells are occupied if r balls are placed at random in n cells, by considering r as a stochastic variable with a Poisson distribution. Several problems can be found in the texts by Karlin (1968, p. 21) and Wilks (1962, p. 189).

In this article we consider the distribution of a stochastic variable having a conditional distribution of a fairly general type where one of the parameters has a priori distribution which is also of a general form. Almost all the cases discussed in the classical problems and many more are then readily obtained as special cases of the general results. Even though the results obtained will also hold good for complex stochastic variables we consider here only the real case. Since there is a sufficiently large number of parameters at our disposal, there is no difficulty in assuming that the special functions, used here to define the probability distributions, are non-negative.

2. Some definitions and preliminary results

The Meijer's G-function is defined by means of Mellin-Barnes type integral as,

$$(1) \quad G_{p,q}^{m,n}(x \mid \begin{smallmatrix} a_1, \dots, a_p \\ b_1, \dots, b_q \end{smallmatrix}) = (1/2 \pi i) \int_L \left\{ \prod_{j=1}^m \Gamma(b_j - s) \cdot \prod_{j=1}^n \Gamma(1 - a_j + s) x^{s/\pi} \prod_{j=m+1}^q \Gamma(1 - b_j + s) \prod_{j=n+1}^p \Gamma(a_j - s) \right\} ds$$

where an empty product is interpreted as unity, $0 \leq m \leq q, 0 \leq n \leq p$ and the contour L is such that it separates the poles of $\Gamma(b_j - s)$ for $j = 1, \dots, n$ from those of $\Gamma(1 - a_j + s)$ for $j = 1, \dots, n$. For the properties, behaviour and particular cases of the G-functions see Erdelyi (1953, pp. 206-222). However, some of the particular cases are given below, which will be used in the investigations that follow.

2.1 Special cases of G-functions

These particular cases are directly related to the classical distributions.

$$(2) \quad x^\beta e^{-px^\beta} = x^\beta G_{0,1}^{1,0}(px^a \mid 0) = p^{-\beta/a} G_{0,1}^{1,0}(px^a \mid \beta/a).$$

This function is associated with the generalized Gamma distribution from which can be derived the Gamma, Chi-square, truncated Normal, Exponential and also the discrete Poisson distributions.

$$(3) \quad x^a (1-x)^\beta = x^a {}_1F_0(-\beta; x) = x^a \Gamma(\beta+1) G_{1,1}^{1,0}(x \mid \beta+1) \\ = \Gamma(\beta+1) G_{1,1}^{1,0}(x \mid \beta+a+1)$$

which is connected with the Beta distribution and the Binomial distribution.

$$(4) \quad x^\beta / (1+a x^a) = x^\beta {}_1F_0(1; ax^a) = x^\beta G_{1,1}^{1,1}(ax^a \mid 0) \\ = a^{-\beta/a} G_{1,1}^{1,1}(a x^a \mid \beta/a)$$

$$(5) \quad {}_pF_q(a_1, \dots, a_p; b_1, \dots, b_q; x) = \frac{\prod_{j=1}^q \Gamma(b_j)}{\prod_{j=1}^p \Gamma(a_j)} G_{p,q+1}^{1,p}(x \mid \begin{matrix} 1-a_1, \dots, 1-a_p \\ 0, 1-b_1, \dots, 1-b_q \end{matrix})$$

for $p \leq q+1$.

$$(6) \quad {}_2F_1(a, b; c; x) = \Gamma(c) G_{2,2}^{1,2}(x \mid \begin{matrix} 1-a, 1-b \\ 0, 1-c \end{matrix}) / \Gamma(a) \Gamma(b).$$

Several statistical distributions are associated with ${}_2F_1(a, b; c; x)$. For a list of them see Mathai and Saxena (1966).

$$(7) \quad {}_1F_1(a; c; x) = \Gamma(c) G_{1,2}^{1,1}(x \mid \begin{matrix} 1-a \\ 0, 1-c \end{matrix}) / \Gamma(a).$$

$$(8) \quad {}_0F_1(c; x) = \Gamma(c) G_{0,2}^{1,0}(x \mid 0, 1-c).$$

$$(9) \quad (1-x)^{a-1} = \Gamma(a) G_{1,1}^{1,0}(x \mid \begin{matrix} a \\ 0 \end{matrix}).$$

$$(10) \quad \Gamma(a) (1-x)^{-a} = G_{1,1}^{1,1}(-x \mid \begin{matrix} 1-a \\ 0 \end{matrix}).$$

$$(11) \quad x^\sigma G_{p,q}^{m,n}(x \mid \begin{matrix} a_1, \dots, a_p \\ b_1, \dots, b_q \end{matrix}) = G_{p,q}^{m,n}(x \mid \begin{matrix} a_1+\sigma, \dots, a_p+\sigma \\ b_1+\sigma, \dots, b_q+\sigma \end{matrix}).$$

3. Gamma with G-functions

We consider here the case where the conditional distribution, assumed to be of a generalized Gamma type, contains a parameter with an a priori G-distribution.

Theorem 1. Let the conditional distribution of a stochastic variable X (discrete or continuous), given the parameter p , be a generalized Gamma with the density function,

$$(12) f(x|p) = \beta p^{a/\beta} x^{a-1} e^{-px^\beta} / \Gamma(a/\beta), \quad 0 < x < \infty \text{ and } f(x|p) = 0$$

elsewhere, where a, β are additional parameters. Let the marginal distribution of p be given by a general density $g(p)$, hereafter called the G -distribution,

$$(13) g(p) = \left\{ \prod_{j=k+1}^s \pi \Gamma(-b_j) \prod_{j=l+1}^r \pi \Gamma(1+a_j) / \prod_{j=1}^k \pi \Gamma(1+b_j) \prod_{j=1}^l \pi \Gamma(-a_j) \right\}$$

$$\lambda G_{r,s}^{k,l}(\lambda p \mid \frac{a_1, \dots, a_r}{b_1, \dots, b_s}), \quad p > 0,$$

where $a_1, \dots, a_r, b_1, \dots, b_s$ are parameters and k, l, r, s are positive integers. Then the unconditional distribution of X is given by the density.

$$(14) h(x) = \left\{ \lambda \beta \prod_{j=k+1}^s \pi \Gamma(-b_j) \prod_{j=l+1}^r \pi \Gamma(1+a_j) / \prod_{j=1}^k \pi \Gamma(1+b_j) \prod_{j=1}^l \pi \Gamma(-a_j) \Gamma(a/\beta) \right\}$$

$$x^{-(\beta+a)} G_{r+l,s}^{k,l+1}(\lambda/x^\beta \mid \frac{-a/\beta, a_1, \dots, a_r}{b_1, \dots, b_s})$$

where $k+l > (r+s)/2$; $R(b_j+1+a/\beta) > 0$, $j = 1, \dots, k$; and $R(\cdot)$ means the real part of $(\cdot)$.

Proof: The unconditional distribution is given by,

$$(15) h(x) = \int_0^\infty f(x|p) g(p) dp$$

This can be evaluated by using the result, that,

$$(16) \int_0^\infty e^{-qt} t^a G_{r,s}^{k,l}[\lambda t \mid \frac{a_1, \dots, a_r}{b_1, \dots, b_s}] dt = q^{-a-1} G_{r+1,s}^{k,l+1}[\lambda/q \mid \frac{-a, a_1, \dots, a_r}{b_1, \dots, b_s}]$$

(Erdelyi, 1954, p. 419). The required result is then obtained by replacing q by x^β , a by a/β and then multiplying by the normalizing factors in $f(x|p)$ and $g(p)$.

3.1 Particular cases

As special cases of $f(x|p)$ we can obtain a number of distributions.

$\beta = 1$: Gamma distribution with parameters a and p . The same function can be taken as the probability function for a Poisson variate with a as the variable taking values $1, 2, \dots$, and with xp as the parameter.

$\beta = 2, p = 1/2\sigma^2$: Truncated Normal distribution with mean zero and with variance σ^2 .

$\beta = 1, \alpha = 1$: Exponential distribution with parameter p .

The following table lists a few particular cases where the conditional distribution is generalized Gamma mentioned above and the a priori distribution for an appropriate parameter is related to one of the special cases mentioned in section 2.1.

$g(p)$	$h(x) = \int_0^{\infty} f(x p)g(p) dp$
(a) Exponential $\lambda e^{-p\lambda} = \lambda G_{0,1}^1(0) (\lambda p 0)$	$\lambda \alpha x^{\alpha-1} (\lambda + x^\beta)^{-\alpha/\beta-1}$ obtained from (14) on using the formulae (11), (2) and (10).
(b) Gamma $p^{\alpha-1} \lambda^\alpha e^{-p\lambda} / \Gamma(\alpha)$	$\lambda^\alpha \beta \Gamma(\alpha + \alpha/\beta) x^{\alpha-1} (\lambda + x^\beta)^{-\alpha(1+1/\beta)} / \Gamma(\alpha) \Gamma(\alpha/\beta)$ obtained from (14) on applying the formulae (2), (11) and (10). These can also be obtained by direct computation.
(c) Beta $\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} p^{\alpha-1}(1-p)^{\beta-1}$	$\beta \Gamma(\alpha+\beta) \Gamma(\alpha+\alpha/\beta) / \Gamma(\alpha/\beta) \Gamma(\alpha)$ $\Gamma(\alpha+\beta+\alpha/\beta)$ ${}_1F_1(\alpha+\alpha/\beta; \alpha+\alpha/\beta+\beta; -x^\beta) x^{\alpha-1}$ obtained from (14) on taking $\lambda = 1$ and using the formulae (9) and (7).

4. A generalized hypergeometric with a G-distribution

From the particular cases it may be noticed that the Binomial distribution is also covered under this case.

Theorem 2. Let a stochastic variable X , (discrete or continuous), have a conditional distribution,

$$(17) f(x|p) = p^{\rho-1} (1-p)^{x-1} {}_2F_1(a, \beta; x; 1-p) / \frac{\Gamma(x)\Gamma(\rho)\Gamma(x+\rho-\alpha-\beta)}{\Gamma(x+\rho-\alpha)\Gamma(x+\rho-\beta)},$$

$0 < p < 1$ and for an appropriate range of values of x , where ρ may be a function of x and $R(\rho) > 0, R(x+\rho-\alpha-\beta) > 0$. Let the marginal distribution of p be given by,

$$(18) g(p) = G_{r,s}^{m,n}(ap | a_1, \dots, a_r; b_1, \dots, b_s) / G_{r+1, s+1}^{m, n+1}(a | 0, a_1, \dots, a_r; b_1, \dots, b_s, -1)$$

where $m + n > (r+s)/2, R(1+b_j) > 0$ for $j = 1, \dots, m$. Then the unconditional distribution of X is given by,

$$(19) \quad h(x) = \frac{\Gamma(x+\rho-a)\Gamma(x+\rho-\beta) G_{r+2,s+2}^{m,n+2}(a \mid 1-\rho, 1-\rho+a+\beta-x, a_1, \dots, a_r; b_1, \dots, b_s, 1+a-x-\rho, 1+\beta-x-\rho)}{\Gamma(\rho) \Gamma(x+\rho-a-\beta) G_{r+1,s+1}^{m,n+1}(a \mid 0, a_1, \dots, a_r; b_1, \dots, b_s, 1)}$$

where $R(\rho+x-a-\beta) > 0$, $R(\rho+x-a-\beta + \min \beta_j) > 0$, $|a| < (m+n-r/2-s/2)$, $m+n > (r+s)/2$.

Proof: The unconditional distribution is given by

$$h(x) = \int_0^1 f(x|p) g(p) dp$$

$$(20) \quad \text{But, } \int_0^1 t^{\rho-1} (1-t)^{\gamma-1} {}_2F_1(a, b; \gamma; 1-t) G_{r,s}^{m,n}(at \mid a_1, \dots, a_r; b_1, \dots, b_s) dt$$

$$= \Gamma(\gamma) G_{r+2,s+2}^{m,n+2}(a \mid 1-\rho, 1-\rho+a+\beta-\gamma, a_1, \dots, a_r; b_1, \dots, b_s, 1+a-\gamma-\rho, 1+\beta-\gamma-\rho)$$

(Erdelyi, 1954, p. 421). The required result is then obtained on multiplying this by the appropriate constants. On using the formulae (11) and (5) we obtain the following corollary.

Corollary. Let a stochastic variable, (discrete or continuous), have a conditional distribution, $f(x|p)$ as defined in (17), and let the marginal distribution of p be,

$$(21) \quad g(p) = p^{\sigma-1} \sigma {}_rF_s(a_1, \dots, a_r; b_1, \dots, b_s; \sigma p) / {}_{r+1}F_{s+1}(\sigma, a_1, \dots, a_r; b_1, \dots, b_s, \sigma+1; a)$$

where $R(\sigma) > 0$. Then the unconditional distribution of X is given by,

$$(22) \quad h(x) = \frac{\sigma \Gamma(\rho+\sigma-1) \Gamma(\rho+\sigma+x-a-\beta-1) \Gamma(x+\rho-\beta) \Gamma(x+\rho-a)}{\Gamma(x+\rho+\sigma-a-1) \Gamma(x+\sigma+\beta+\rho-1) \Gamma(\rho) \Gamma(x+\rho-a-\beta)}$$

$${}_{r+2}F_{s+2}(\rho+\sigma-1, x+\rho-a-\beta+\sigma-1, a_1, \dots, a_r; x+\rho+\sigma-\beta-1, x+\rho+\sigma-a-1, b_1, \dots, b_s; a) /$$

$${}_{r+1}F_{s+1}(\sigma, a_1, \dots, a_r; b_1, \dots, b_s, \sigma+1; a)$$

where $R(\rho+\sigma) > 1$, $R(\rho+x+\sigma-a-\beta) > 1$.

4.1 Particular cases

In the following table we give the unconditional distribution of a stochastic variable X which has a conditional distribution,

$$(23) f(x|p) = p^{\rho-1} (1-p)^{x-1} / \{\Gamma(x) \Gamma(\rho) / \Gamma(x+\rho)\}.$$

This is obtained by putting $\alpha = 0$ and $\beta = 0$ in (17). It may be noticed that if x is replaced by $x+1$ and ρ by $n-x+1$ then we get the Binomial distribution. The marginal distributions are given in the following table.

$g(p)$	$h(x) = \int_0^1 f(x p) g(p) dp$
(a) $\frac{\Gamma(\sigma+k) p^{\sigma-1} (1-p)^{k-1}}{\Gamma(\sigma)\Gamma(k)}$	$\frac{\sigma\Gamma(\rho+\sigma-1)\Gamma(x+\rho)\Gamma(x+k-1)\Gamma(k+\sigma)}{\Gamma(x)\Gamma(\sigma+1)\Gamma(k)\Gamma(\rho)\Gamma(x+\rho+\sigma+k-2)}$
This is obtained by putting $a = 0$, $p = 1$, $s = 0$, $a_1 = 1-k$ in (21), then using the result, ${}_2F_1(a, b, c, 1) = \frac{\Gamma(c)\Gamma(c-a-b)}{\Gamma(c-a)\Gamma(c-b)}$ where, $R(c) > 0$, $R(c-a-b) > 0$ and ${}_1F_0(\lambda; z) = (1-z)^{-\lambda}$	where $R(\rho+\sigma) > 1$, $R(x+k-1) > 0$.
(b) $p^{\sigma-1} \sigma e^{-ap} / {}_1F_1(\sigma; \sigma+1; -a)$	$\frac{\sigma\Gamma(\rho+\sigma-1)\Gamma(x+\rho) {}_1F_1(\rho+\sigma-1; x+\rho+1; -a)}{\Gamma(x+\rho+\sigma-1)\Gamma(\rho) {}_1F_1(\sigma; \sigma+1; -a)}$
This is obtained by putting $r = s = 0$ in (21).	where $R(\rho+\sigma) > 1$.
(c) $\frac{p^{\sigma-1} \sigma_0 F_1(-; v; ap)}{{}_1F_2(\sigma; \sigma+1, v; a)}$	$\frac{\sigma\Gamma(\rho+\sigma-1)\Gamma(x+\rho) {}_1F_2(\rho+\sigma-1; x+\rho+1, v; a)}{\Gamma(x+\rho+\sigma-1)\Gamma(\rho) {}_1F_2(\sigma; \sigma+1, v; a)}$
	where $R(\rho+\sigma) > 1$.

Remarks. Several such combinations of the special cases can be easily obtained by specializing the parameters.

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